

AI CRM Lead Automation

Python 3.11+ local application | Standard-library runtime | Optional OpenAI Responses API | MIT licensed

DELIVERY	ANALYSIS	QUALITY	DEPLOYMENT
Local dashboard + API	Offline or optional OpenAI	50 automated tests	Local-first application

Architecture

INTAKE	ANALYSIS	PERSISTENCE	DELIVERY
Threaded local HTTP server, strict JSON media type, request/message bounds, normalized object schema.	Versioned scoring policy or PII-redacted Structured Outputs; local validation always enforced.	Process lock, duplicate check, versioned transactional state, raw message opt-in only.	Escaped dashboard, JSON endpoints, metrics, event log, formula-safe CSV.

HTTP surface

METHOD	ROUTE	PURPOSE
GET	/ and /api/health	Dashboard and mode-aware health check
POST	/api/intake	Validate, classify, score, and persist one lead
GET	/api/leads and /api/logs	Structured CRM records, metrics, and audit events
GET	/export/leads.csv	Formula-neutralized spreadsheet handoff

Security and privacy controls

- Loopback-only host validation	- HTML and CSS output constraints
- Bounded request body and message	- CSV formula neutralization
- Safe client-facing errors	- No secrets or lead text in logs
- Security and no-cache headers	- Raw inquiry retention off by default

Verification

50 isolated tests cover configuration, startup, seed validation, scoring, PII redaction, strict provider output, retries, transactional storage, concurrency, CSV and HTML safety, and local HTTP behavior. CI enforces branch coverage on Python 3.11-3.13, verifies the balanced scoring set, and rebuilds both PDFs.

Run locally

```
python3 -m unittest discover -s tests -v      python3 src/seed_data.py      python3 src/app.py
```

Production rollout includes authentication, TLS, rate limits, managed persistence, monitoring, backups, and formal data lifecycle controls.